RL Assignment-2

18/08/2026

Calculations for Two plays:-

$$Q_{new}(a) = Q_{old}(a) + \frac{R - Q_{old}(a)}{N(a)}$$

Play 1

Selected machine : A

Reward optimined : 3

Initially:

$$Q_{old}(A) = 0, \quad N(A) = 0$$

After selecting A:-

$$N(A) = 0 + 1 = 1$$

Now calculate Q:

$$Q(A) = 0 + \frac{3 - 0}{1}$$

$$Q(A) = 3$$

Therefore, after play 1

$$N(A) = 1, \quad R(A) = 3, \quad Q(A) = 3$$

Play 3:-

Selected machine : C

Reward obtained : 3

Before Play 3:

$$Q_{old}(C) = 4$$

Because C → recurred reward 4 in Play 2

No. of previous selection :

$$N(C) = 1$$

After selecting C again:

$N(C) = 1+1 = 2$

Now calculate Q:

$$Q(C) = 4 + \frac{3-4}{2}$$

$$Q(C) = 4 + \frac{-1}{2}$$

$$\boxed{Q(C) = 3.5}$$

Therefore,

After play 3:

$N(C) = 2$, $R(C) = 7$, $Q(C) = 3.5$

For play 5,

C was selected and reward = 4

Before play 5:

$Q_{old}(C) = 3.3333$

After selection:

$N(C) = 4$

Therefore:

$$Q(C) = 3.3333 + \frac{4 - 3.3333}{4}$$

$$= 3.3333 + 0.1667$$

$$\boxed{Q(C) = 3.5}$$

Analysis:

The ε-greedy strategy balances exploration & explaitation. With $\varepsilon = 0.2$, there is a 20% probability of exploring a randomly selected machine and an 80% probability of exploiting the

machine with the highest Q-value. The Q-values are updated after every play using the incremental simple average method. In this, machine C is selected frequently because it had the highest estimated Q-value during most exploitation steps.